Low-Cost Pressure Sensors for Transient Monitoring

L. Fixture M. Fixture

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FICTIONAL TEST MATERIAL: this paper does not exist; it was written for the citation-audit fixture. No author, utility, venue, or measurement described here is real.

Abstract

Pressure transients in water mains last a fraction of a second, which is far shorter than the sampling interval of a typical network pressure logger. We built low-cost pressure sensors that sample at 100 Hz and installed them beside standard 1 Hz loggers at eight sites of a fictional network. The 100 Hz sensors captured pressure transients above 5 m head that the 1 Hz loggers missed. Each sensor cost a small fraction of a commercial transient logger. We describe the hardware, the side-by-side trial, and the transient events recorded.

1 Introduction

A pressure transient is a short pressure wave caused by a sudden change in flow, such as a pump trip or a fast valve closure. Large or repeated transients fatigue pipes and joints. Standard network loggers record pressure once per second or less often, so a transient that rises and falls within a second can pass between two samples and leave no trace.

Commercial transient loggers sample fast enough but are too expensive to deploy widely. This paper asks whether a low-cost sensor sampling at 100 Hz can capture the transients that the standard loggers miss.

2 Method

2.1 Sensor design

Each sensor combines an industrial pressure transducer, a small microcontroller, and a battery. The microcontroller samples the transducer at 100 Hz and stores any one-second window in which the pressure changes by more than 2 m head. The parts for one sensor cost less than one tenth of the price of a commercial transient logger.

2.2 Side-by-side trial

At eight sites of the fictional Fixture Meadows network, a 100 Hz sensor was installed on the same hydrant tapping as an existing 1 Hz logger. Both devices ran for twelve weeks. Their clocks were synchronised, so each event recorded by one device could be checked against the record of the other.

3 Results

3.1 Transients captured

The low-cost sensors sampling at 100 Hz captured pressure transients above 5 m head that the 1 Hz loggers missed. Over the twelve weeks, the 100 Hz sensors recorded 37 transients larger than 5 m head across the eight sites. The 1 Hz loggers showed no excursion above 5 m head for 31 of these 37 events; for the other six, the 1 Hz record showed only a small step.

3.2 Sources of the transients

Most of the transients coincided with pump starts and stops at a booster station. The others matched fast hydrant closures during flushing work.

3.3 Cost and reliability

All eight sensors ran for the full twelve weeks on one battery. At their low cost, a utility could deploy them at many more sites than it could with commercial transient loggers.

4 Limitations

This study tests transient capture only.

This study does not examine bursts: we did not relate the recorded transients to main breaks, and no burst occurred at the trial sites during the twelve weeks. This study does not examine metering: the sensors measure pressure, not flow, and no flow or consumption data were used. The trial covered eight sites for twelve weeks, so the number of transients is small.